

LYRA: GENERATIVE 3D SCENE RECONSTRUCTION VIA VIDEO DIFFUSION MODEL SELF-DISTILLATION

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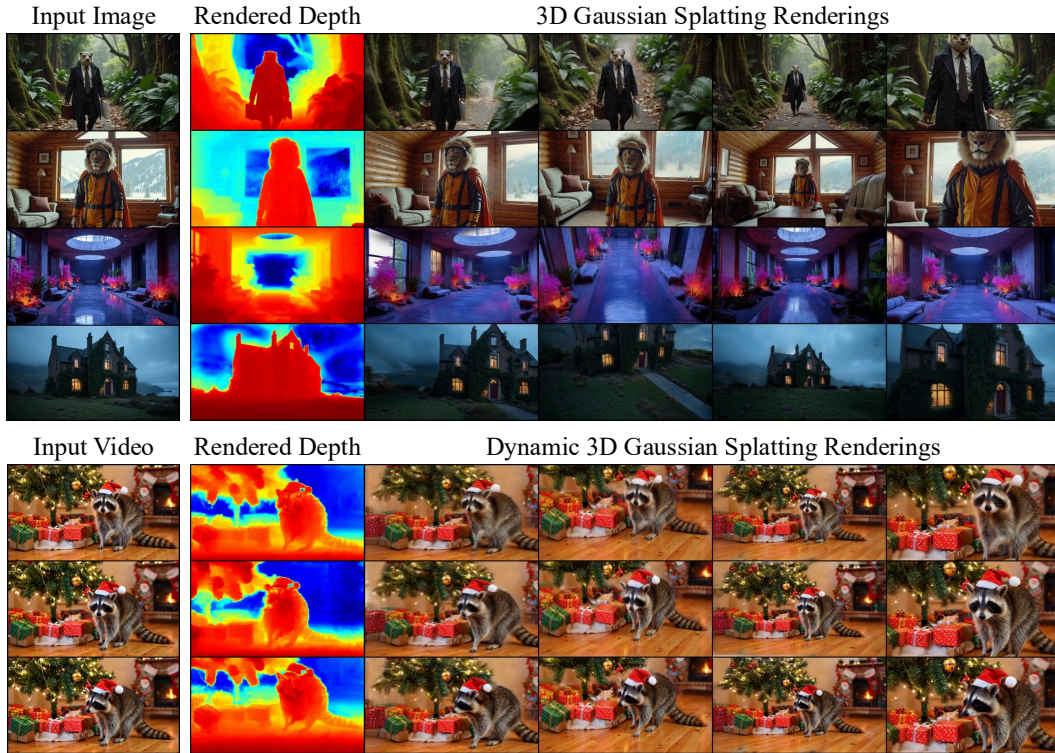


Figure 1: **Feed-Forward 3D and 4D Scene Generation.** From a single image (top), *Lyra* infers a 3D Gaussian Splatting (3DGS) representation in a feed-forward fashion, through self-distilling a video diffusion model without requiring real-world multi-view data. With a video input (bottom), *Lyra* infers a dynamic 3DGS that offers interactive control in both time (rows) and viewpoint (columns).

ABSTRACT

Creating virtual environments as explicit 3D assets is a central requirement for gaming, simulation, robotics, autonomous driving, and industrial AI. Existing learning-based 3D reconstruction systems, however, generally depend on captured real-world multi-view data, a requirement that sharply limits both scale and scene diversity. Video diffusion models offer a complementary supervisory signal: they exhibit broad scene imagination and high visual fidelity, yet their outputs remain two-dimensional and therefore cannot directly serve simulation settings in which an embodied agent must navigate and interact within a geometrically coherent world. We introduce a self-distillation framework that transfers the implicit 3D knowledge of camera-controlled video diffusion models into an explicit 3D Gaussian Splatting (3DGS) representation, eliminating the need for multi-view training captures. The framework augments the standard RGB decoder with a trainable 3DGS decoder, supervised by the RGB decoder’s own generated outputs. As a result, the 3DGS decoder is trained entirely from synthetic supervision produced by video diffusion models. At inference time, the resulting model synthesizes real-time renderable 3D scenes from either a text prompt or a single image. The same self-distillation

principle also supports dynamic 3D scene generation from a monocular input video. Across static and dynamic 3D scene generation evaluations, the proposed framework achieves state-of-the-art performance.

1 INTRODUCTION

Scalable generation of high-quality 3D environments remains a central challenge in computer vision and graphics, with direct consequences for film-making, VR/AR, and physical AI systems evaluated in closed-loop simulation. These applications require explicit 3D representations that render in real time, maintain multi-view consistency, and support downstream interaction with the generated environment.

Recent neural reconstruction methods (Mildenhall et al., 2020; Kerbl et al., 2023; Moenne-Loccoz et al., 2024) recover such representations from posed image collections, but their reliance on accurate cameras and high-quality multi-view imagery constrains scalability. The constraint is even stronger for dynamic scenes, where reconstruction commonly depends on synchronized multi-camera acquisition (Li et al., 2022). Reconstruction is also bounded by the observed evidence and therefore cannot reliably synthesize content outside captured views. Feed-forward reconstruction models (Zhang et al., 2024e; Ren et al., 2024c) reduce per-scene optimization costs, but they expose a different bottleneck: diverse, large-scale 3D supervision remains scarce, limiting generalization beyond the training distribution.

Video diffusion models (Cosmos, 2025; Wan, 2025) provide a powerful source of supervision for overcoming this data limitation. Trained on large video corpora, they deliver high visual fidelity and broad scene coverage; by observing real videos with varied camera motion, including handheld trajectories, cinematic pans, and aerial footage, they implicitly acquire cues about 3D structure without requiring explicit multi-view reconstruction datasets. Unlike purely reconstructive approaches (Charatan et al., 2024), generative video models can synthesize plausible content outside the visible input. Their limitation is therefore representational rather than generative: they output 2D frame sequences rather than explicit 3D assets, preventing direct use in simulation and other settings that require geometric consistency, temporal coherence, and physical interaction. Recent camera-control methods (Wang et al., 2024e; Ren et al., 2025) further show that video models can be conditioned on camera trajectories using relatively modest posed data, turning generated, camera-indexed image sequences into an effective supervisory substrate for 3D representation learning.

We build on this observation and formulate *generative 3D scene reconstruction*: the direct conversion of video-diffusion latents into explicit 3D environments. We introduce *Lyra*, a feed-forward framework that generates 3D Gaussian Splatting (3DGS) scenes from the latent representations of a camera-controlled video diffusion model. The key mechanism is self-distillation (Fig. 2). Given a single image or video, *Lyra* samples a camera trajectory, denoises the corresponding video latent, and decodes that latent through two coordinated branches. The frozen RGB decoder generates a video sequence and acts as the teacher, while the trainable 3DGS decoder operates in the same latent space and produces the explicit 3D student representation. To expand viewpoint coverage, we sample multiple camera trajectories, generate multiple video latents, and train the 3DGS decoder to fuse their information while reducing long-range and multi-view inconsistencies.

This design delivers three central advantages. **First**, it uses camera-controlled video diffusion models to synthesize large-scale, diverse supervision, eliminating dependence on captured real-world multi-view training data. **Second**, because reconstruction is performed in the compressed latent space of the video model, *Lyra* aggregates many high-resolution views without the prohibitive memory cost of pixel-space feed-forward reconstruction. **Third**, the explicit 3DGS output provides a geometrically grounded representation for real-time rendering and downstream simulation. Consequently, at inference time, *Lyra* produces high-quality 3D Gaussian scenes from monocular input *without requiring any additional optimization or post-processing*.

We further extend the same self-distillation principle to dynamic 4D generation from monocular video. In this setting, the video model supplies space-time supervision, and the student learns time-conditioned 3DGS representations that support novel-view synthesis of dynamic scenes.

Overall, our work contributes the following:

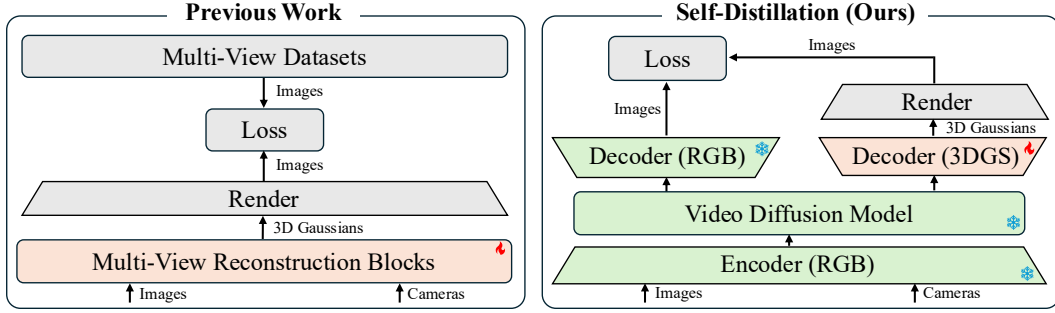


Figure 2: **Self-distillation framework of *Lyra***. Previous work (left) (Szymanowicz et al., 2025b) trains multi-view reconstruction blocks using real-world datasets with limited diversity (Zhou et al., 2018; Ling et al., 2024b). In contrast, we propose a self-distillation framework (right) for generative 3D scene reconstruction. Precisely, a pre-trained camera-controlled video diffusion model with its RGB decoder output (teacher) supervises the rendering of the 3DGS decoder (student). Using a video model pre-trained on diverse 2D video data allows us to provide diverse multi-view supervision.

- A self-distillation framework that trains a 3DGS student decoder from a pre-trained camera-controlled video diffusion teacher, removing the requirement for captured real-world multi-view data.
- A latent-space feed-forward reconstruction formulation that aggregates many generated high-resolution views while avoiding the memory limits of pixel-space reconstruction.
- A dynamic extension that applies the same formulation to feed-forward 4D reconstruction from a monocular video.
- Empirical evidence of strong generalization across diverse scenes, with state-of-the-art results in single-image 3D scene generation and single-video 4D scene generation.

2 RELATED WORK

Multi-view image generation. Early multi-view image generation methods (Watson et al., 2023; Liu et al., 2023; Shi et al., 2024; Wang & Shi, 2023) primarily target object-centric settings and typically omit scene-scale background structure. Subsequent work (Sargent et al., 2024; Tang et al., 2023b; Schneider et al., 2025) broadens the paradigm to scene-scale generation. CAT3D (Gao et al., 2024b), for example, adapts a pre-trained image diffusion model to generate multi-view images from either generated or real image inputs; a separate reconstruction stage then converts those generated views into an explicit NeRF (Mildenhall et al., 2020) or 3DGS (Kerbl et al., 2023) representation. Related space-time diffusion models extend this pipeline to dynamic 3D scenes observed from multiple viewpoints (Watson et al., 2024; Wu et al., 2025b; Kuang et al., 2024; Wang et al., 2025a). Although these methods produce compelling multi-view imagery, their explicit 3D grounding depends on an expensive per-scene optimization stage that is not amortized across generated scenes. Our objective is instead to generate an explicit 3D scene *directly* from text or a single image.

Camera-conditioned video generation. Camera control has become a defining capability for video diffusion models. MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models with camera poses. Follow-up methods (He et al., 2024a; Xu et al., 2024c; Bahmani et al., 2025a;b) use Plücker-coordinate representations for pixel-wise camera conditioning, while another line of work (Hu et al., 2024; Hou et al., 2024; Zhang et al., 2024b) studies training-free camera control. More recently, SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a) have demonstrated high-quality camera-controlled video synthesis using large-scale synthetic training data. These systems provide camera-aware 2D sequences but do not themselves produce explicit 3D structure. Our work is complementary: we use a camera-controlled video model as the teacher for feed-forward 3D generation, so that novel-view rendering remains possible *after* generation. Concretely, we build on GEN3C (Ren et al., 2025) and distill its camera-controlled video outputs into an explicit 3DGS decoder.

Feed-forward 3D models. Early feed-forward 3D reconstruction methods largely focus on object-centric scenes without backgrounds, directly predicting NeRF (Hong et al., 2024; Li et al., 2024b) or 3DGS (Tang et al., 2024a; Szymanowicz et al., 2024) representations from text or images. Recent scene-level systems (Charatan et al., 2024; Zhang et al., 2024e; Szymanowicz et al., 2025a; Ren et al., 2024c) typically regress per-pixel 3D Gaussians from one or more input images. Because these

models are commonly trained on datasets such as RealEstate10K (Zhou et al., 2018), they can remain tied to training-distribution scenes and may generalize poorly to diverse generated environments. Several methods therefore combine feed-forward reconstruction with generative scene synthesis. Bolt3D (Szymanowicz et al., 2025b) trains a pointmap (Wang et al., 2024b) autoencoder and generates pointmaps alongside multi-view images for feed-forward 3D reconstruction. Wonderland (Liang et al., 2025a), which is closely related to our work, uses a camera-controlled video model and a feed-forward network to generate 3D Gaussians. In contrast, we train the 3D student through self-distillation from a camera-controlled video teacher *without requiring real-world multi-view data*. We further show that the same formulation extends to feed-forward 4D scene generation, a setting not addressed by these prior systems.

3 SELF-DISTILLATION USING VIDEO DIFFUSION MODELS

Our key idea is to distill the implicit 3D knowledge embedded in video diffusion models into an explicit 3DGS decoder capable of producing high-quality 3D representations. To this end, we build a teacher–student framework in which the video diffusion model (teacher) generates RGB videos that supervise the 3DGS decoder (student) that operates in the same latent space, as illustrated in Fig. 2. In this section, we detail our video diffusion model backbone (Sec. 3.1) and its role in self-distillation for the 3DGS decoder (Sec. 3.2). We discuss the design choice for the 3DGS decoder in Sec. 4 and outline minimal changes required to adapt the pipeline to dynamic 3D scenes in Sec. 5.

3.1 BACKGROUND: CAMERA-CONTROLLED AND 3D-CONSISTENT VIDEO DIFFUSION

Since we supervise our 3DGS decoder only with the camera-controlled video diffusion model, we heavily rely on its 3D consistency. Due to this, we build our approach on GEN3C (Ren et al., 2025), a recent camera-conditioned video diffusion model. In the following we provide an overview of the key design choices made in GEN3C.

Spatiotemporal 3D cache. To improve video consistency and camera control precision, GEN3C constructs a spatiotemporal 3D cache $\{\mathbf{P}^{t,v}\}$ from input image(s) or videos, where each $\mathbf{P}^{t,v}$ is a colored point cloud obtained by unprojecting the depth estimation (Wang et al., 2024a) of an RGB image captured from camera viewpoint v at time t . The cache is organized as an $L \times V$ array, where L denotes the number of frames (temporal length) and V denotes the number of camera views.

Rendering and structured guidance. To leverage this cache, GEN3C renders each point cloud from arbitrary given camera poses, producing RGB images $\mathbf{I}^{t,v}$ and disocclusion masks $\mathbf{M}^{t,v}$ via $(\mathbf{I}^{t,v}, \mathbf{M}^{t,v}) = \mathcal{R}(\mathbf{P}^{t,v}, \mathbf{C}^t)$, where $\mathcal{R}$ denotes the rendering function that projects the 3D point cloud $\mathbf{P}^{t,v}$ onto the 2D camera plane according to the camera pose $\mathbf{C}^t$ at time t . Given a sequence of camera poses $\mathbf{C} = (\mathbf{C}^1, \dots, \mathbf{C}^L)$, rendering all cache elements produces V videos of length L , which can be stacked into image sequences $\mathbf{I}^v \in \mathbb{R}^{L \times 3 \times H \times W}$ and mask sequences $\mathbf{M}^v \in \mathbb{R}^{L \times 1 \times H \times W}$ for each view v . The disocclusion masks indicate areas that the video diffusion model should fill in. These renderings serve as structured visual guidance for subsequent video generation.

Video variational autoencoder (VAE). In diffusion-based video generation models, a video variational autoencoder (VAE) (Kingma & Welling, 2013; Rombach et al., 2022) is commonly employed to compress videos into a lower-dimensional latent space for efficient training and inference. Given a RGB video $\mathbf{I} \in \mathbb{R}^{L \times 3 \times H \times W}$, a pre-trained VAE encoder $\mathcal{E}$ will encode the video into a latent space, i.e. $\mathbf{z} = \mathcal{E}(\mathbf{I}) \in \mathbb{R}^{L' \times C \times h \times w}$. The training and inference of the diffusion model are then performed in this latent space. The final video $\hat{\mathbf{I}} = \mathcal{D}_{rgb}(\mathbf{z})$ is decoded with a pre-trained VAE decoder $\mathcal{D}_{rgb}$. In this paper, we adopt the pre-trained GEN3C (Ren et al., 2025; Cosmos, 2025) model. Specifically, the latent channel dimension is $C = 16$, the temporal dimension is $L' = (L - 1)/\tau + 1$, and the spatial dimensions are $h = H/\sigma$ and $w = W/\sigma$, where the temporal compression factor is $\tau = 8$ and the spatial compression factor is $\sigma = 8$.

3.2 SELF-DISTILLATION

We train the 3DGS decoder $\mathcal{D}_s$ as a student under a teacher–student paradigm, where a camera-controlled video diffusion model $\mathcal{V}$ serves as the teacher. For diverse supervision, we curate a large-scale set of diverse text prompts with large language models (OpenAI, 2025; Bai et al., 2023),

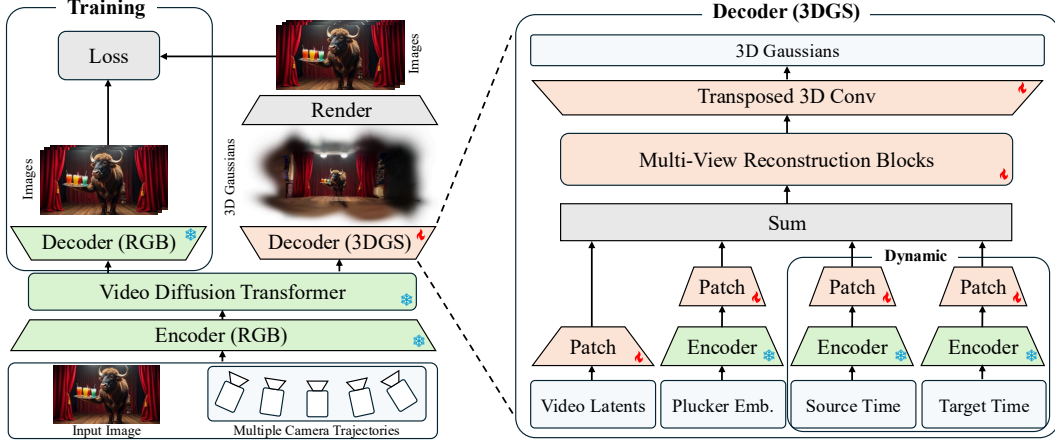


Figure 4: **3D Generative Reconstruction Framework.** Our pipeline builds upon a camera-controlled video diffusion model (Ren et al., 2025) pre-trained on large scale data. We train a 3D Gaussian Splatting (3DGS) decoder by aligning the 2D image renderings of generated 3DGS scenes with the RGB-decoded generations of the pre-trained video model. We only train the 3DGS decoder while freezing the pre-trained autoencoder and diffusion model. At inference time we directly use the 3DGS decoder, without requiring the RGB decoder anymore. Time conditioning within the 3DGS decoder allows us to easily extend our approach from static to dynamic 3D scene generation.

generate images I with an image diffusion model (Black Forest Labs, 2024), and expand into multi-view sequences using GEN3C (Ren et al., 2025).

Teacher–student setup. Given an input image I and a sampled camera trajectory $\{\mathbf{C}^t\}_{t=1}^L$, the video diffusion model $\mathcal{V}$ generates a denoised video latent $\mathbf{z} = \mathcal{V}(I, \{\mathbf{C}^t\}_{t=1}^L)$. The latent $\mathbf{z}$ is decoded along two branches: the pre-trained RGB decoder $\mathcal{D}_{rgb}$ produces video frames $\mathbf{I}_{\mathcal{D}_{rgb}} = \mathcal{D}_{rgb}(\mathbf{z})$, while the 3DGS decoder $\mathcal{D}_s$ outputs explicit 3D Gaussians $\mathbf{G}$. Rendered views from 3DGS (*student*) are defined as $\mathbf{I}_{\mathcal{D}_s} = \text{Render}(\mathbf{G}, \{\mathbf{C}^t\}_{t=1}^L)$ and are supervised to match RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}$ (*teacher*), forming the self-distillation loop.

Multi-trajectory supervision. In our paper, to enlarge viewpoint coverage for the input image, we sample $V = 6$ camera trajectories per input image, as shown in Fig. 3. For each trajectory v , we construct a spatiotemporal cache $\{\mathbf{P}^{t,v}\}_{t=1}^L$ along with $L = 121$ camera poses $\{\mathbf{C}^{t,v}\}_{t=1}^L$. Passing these caches through video diffusion model $\mathcal{V}$ yields latents $\mathbf{z}^v$, which the teacher decodes into RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}^v = \mathcal{D}_{rgb}(\mathbf{z}^v)$. The 3DGS decoder $\mathcal{D}_s$ learns to fuse these multiple $\mathbf{z}^v$ into coherent Gaussians $\mathbf{G}$, while filling disoccluded regions. Through this self-distillation design, the 3DGS decoder $\mathcal{D}_s$ is trained entirely from synthetic supervision provided by $\mathcal{V}$. Operating in latent space enables efficient aggregation of multiple trajectories, while the explicit Gaussian output $\mathbf{G}$ enforces geometric consistency and supports downstream simulation and real-time rendering.

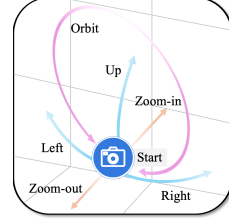


Figure 3: Sampled six camera trajectories to maximize view coverage.

4 FEED-FORWARD RECONSTRUCTION FROM MULTI-VIEW VIDEO LATENTS

Our feed-forward 3DGS decoder $\mathcal{D}_s$ is designed to transform the synthesized multi-view latents generated by our video model $\mathcal{V}$ into an explicit 3D representation that can be rendered from arbitrary viewpoints. We choose 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) due to its explicit representation and fast rendering speed. We visualize our detailed 3DGS decoder in Fig. 4 (right).

4.1 3DGS DECODER

Scaling Latent-Based 3D Reconstruction. Previous feed-forward reconstruction frameworks generate 3D Gaussians for each pixel, and thus, are limited by the number and resolution of input views they can handle. For example, GS-LRM (Zhang et al., 2024e) operates on 2–4 images at 512×512 resolution, while AnySplat (Jiang et al., 2025a) is trained with 24 images at 448×448

resolution. Our video model synthesizes $V \times L = 6 \times 121 = 726$ input views at 704×1280 resolution—far beyond the capacity of existing methods, which exceeds GPU memory limit. The bottleneck lies in the attention mechanism applied to visual tokens, whose memory and compute requirements grow with the number of pixels. To overcome this limitation, we avoid scaling in pixel space and instead operate directly in the compressed video latent space produced by our camera-controlled video diffusion model. The multi-view video latents are denoted as $\mathbf{Z} \in \mathbb{R}^{V \times L' \times C \times h \times w}$.

Architecture. The 3DGS decoder $\mathcal{D}_s$ first maps its inputs into the hidden dimension of the main reconstruction blocks and outputs per-pixel 3D Gaussian features $\mathbf{G} \in \mathbb{R}^{V \times L \times H \times W \times 14}$. Specifically, the inputs are the multi-view video latents $\mathbf{Z}$ and encoded Plücker embeddings $\mathbf{E}$, both with latent dimensions $\mathbf{Z}, \mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. Each input component is first patchified to match the hidden dimension of the main reconstruction blocks. We do not require any additional visual encoder—only a spatial 2×2 patchification layer (Dosovitskiy et al., 2021) to transform the video latents into flattened tokens for the reconstruction network. The sum of both inputs is processed through the reconstruction blocks. We follow the design introduced in Long-LRM (Ziwen et al., 2024), i.e., one block consists of one Transformer (Vaswani et al., 2017) layer followed by seven Mamba-2 (Dao & Gu, 2024) layers. We repeat the block twice to get 16 layers with 512 hidden dimensions. Finally, a transposed 3D convolution maps the hidden representation to 14 Gaussian channels: 3D position (x, y, z) , scale (s_x, s_y, s_z) , rotation quaternion (q_w, q_x, q_y, q_z) , opacity α , and RGB (r, g, b) . Formally, the static decoder is expressed as $\mathbf{G} = \mathcal{D}_s(\mathbf{Z}, \mathbf{E})$.

Plücker embeddings. Raw Plücker embeddings are first computed from camera poses $\{\mathbf{C}^{t,v}\}$ and intrinsics as $\mathbf{E}^{\text{raw}} \in \mathbb{R}^{V \times L \times 6 \times H \times W}$. We reuse the RGB encoder $\mathcal{E}$ from the pre-trained VAE to encode Plücker embeddings. Specifically, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins using $\mathcal{E}$, and concatenate latent along the channel dimension to obtain $\mathbf{E}^{\text{enc}} \in \mathbb{R}^{V \times L' \times 2C \times h \times w}$. A lightweight MLP maps $\mathbf{E}^{\text{enc}}$ to $\mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$, reducing the channel dimension by a factor of 2 to match video latent $\mathbf{Z}$.

4.2 LOSS FUNCTION

Image-based supervision. We supervise our 3DGS decoder with an image-based reconstruction loss. The reconstruction loss is split into a Mean Squared Error (MSE) loss $\mathcal{L}_{mse}$ and an LPIPS loss $\mathcal{L}_{lpips}$ (Zhang et al., 2018) using VGG (Simonyan & Zisserman, 2014) as a feature extractor.

Depth supervision. We observed that the decoder trained with only RGB loss often produces flattened geometry. To address this, we additionally supervise the rendered depth maps using the consistent video depth estimated by an off-the-shelf system ViPE (Huang et al., 2025), and the scale-invariant depth loss $\mathcal{L}_{depth}$ from Long-LRM (Ziwen et al., 2024).

Opacity-based pruning. Similar to Long-LRM (Ziwen et al., 2024), we use an L1 regularization on the opacity $\mathcal{L}_{opacity}$ and remove the Gaussians with the lowest 80% opacity.

Total loss. Our total loss is computed as the weighted sum of all losses with weight factors λ_i

$$\mathcal{L} = \lambda_{mse} \mathcal{L}_{mse} + \lambda_{lpips} \mathcal{L}_{lpips} + \lambda_{depth} \mathcal{L}_{depth} + \lambda_{opacity} \mathcal{L}_{opacity} \quad (1)$$

where we set $\lambda_{mse} = 1.0$, $\lambda_{lpips} = 0.5$, $\lambda_{depth} = 0.05$, and $\lambda_{opacity} = 0.1$.

5 EXTENSION TO DYNAMIC 3D SCENES

Our approach can be extended to handle dynamic 3D scenes with minimal changes, outlined below.

Self-distillation for dynamic scenes. Our dynamic 3D setup closely follows the design of the static 3D counterpart. Instead of a single image, the video model takes a single video of length L with corresponding camera poses as input and generates multi-view video latents capturing the same underlying motion. For video inputs, we follow a protocol similar to the static case: sampling diverse text prompts with large language models, then generating videos using video diffusion models (Cosmos, 2025; Wan, 2025). The generated videos are then annotated with camera poses and depth maps using ViPE (Huang et al., 2025).

Dynamic 3DGS decoder architecture. We follow the bullet-time design of Liang et al. (2025b) and design our dynamic 3DGS decoder $\mathcal{D}_s$ to generate time-dependent 3D Gaussians for all the video

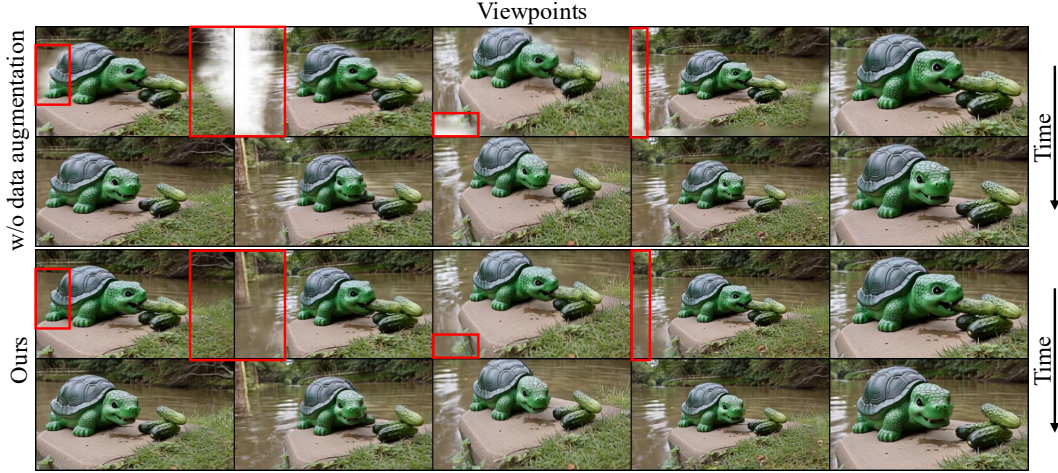


Figure 5: **Dynamic data augmentation.** When naively supervising the time-aware 3DGS decoder, we observe artifacts in the generated 3D Gaussians. Specifically, early timesteps from extreme poses exhibit low opacity in regions not covered by the supervision signal. To address this, we augment the supervision data with a motion-reversed video, ensuring that each timestep is observed from the full spatial coverage, thereby preventing low opacity artifacts in the early timesteps.

frames. Specifically, the dynamic 3DGS decoder $\mathcal{D}_s$ closely follows the architecture of $\mathcal{D}_s$, except that we augment the $\mathcal{D}_s$ input with the encoded source and target time embeddings $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The unencoded (raw) source times are assigned to each input frame, forming $\mathbf{T}^{\text{src, raw}} \in \mathbb{R}^{V \times L \times 1 \times H \times W}$, and the target time is $\mathbf{T}^{\text{tgt, raw}} \in \mathbb{R}^{V \times 1 \times 1 \times H \times W}$. Both are spatially repeated and augmented with a 2-dimensional sinusoidal embedding to expand the channel dimension to 3, then encoded with the RGB encoder $\mathcal{E}$ to latent dimensions $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The target time is repeated along the latent temporal dimension to match L' . During training, a target time step is randomly sampled, and the corresponding 3D Gaussians are decoded and supervised across all trajectories generated with dynamic data augmentation. We fine-tune $\mathcal{D}_d$ from a pre-trained $\mathcal{D}_s$ with the patchification layers of $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ initialized to zeros. $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ are directly added to the sum of $\mathbf{Z}$ and $\mathbf{E}$. Formally, the dynamic decoder is defined as $\mathbf{G} = \mathcal{D}_d(\mathbf{Z}, \mathbf{E}, \mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}})$.

Dynamic data augmentation. In the static 3D scenes, where time is not a factor, frames from different timesteps can all be used to supervise the 3DGS renderings. In contrast, for dynamic 3D scenes, the 3DGS changes over time, so only frames from the corresponding timestamp are valid for supervision. This restriction can lead to a trivial solution, where the 3DGS at a given timestep simply ignores information from frames at other timesteps (see Fig. 5).

To mitigate this issue, we introduce a dynamic data augmentation strategy that balances supervision across timesteps. Early frames are naturally associated with viewpoints close to the input image, whereas later frames correspond to more distant viewpoints. To counter this imbalance, we augment the training data with paired supervision views for each timestep — one near and one far. Concretely, we reverse the input video in frame order and feed it into the video model $\mathcal{V}$, which produces six additional multi-view sequences. After reversing these sequences back to the original motion order, we obtain six trajectories that progress inward (from far viewpoints toward the input). Combined with the original six outward trajectories, this yields 12 supervision views per timestep. Importantly, this augmentation is applied only during training: the reversed trajectories act purely as extra supervision signals to prevent collapse from pruning artifacts and are not required at inference time.

6 EXPERIMENTS

6.1 EXPERIMENTAL SETUP

Datasets. To train our model, we do not use any existing multi-view datasets; instead, we construct our own, which we call the *Lyra* dataset, relying solely on our video model to supervise the 3DGS decoder. The 3D reconstruction setup uses 59,031 images, while the 4D setup has 7,378 videos. All data are from diverse text prompts, spanning scenarios such as indoor and outdoor environments,

Table 1: **State-of-the-art comparisons.** We compare our method with previous works for single image to 3D generation using RealEstate10K, DL3DV, and Tanks-and-Temples datasets.

Method	RealEstate10K			DL3DV			Tanks-and-Temples		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
ZeroNVS	13.01	0.378	0.448	13.35	0.339	0.465	12.94	0.325	0.470
ViewCrafter	16.84	0.514	0.341	15.53	0.525	0.352	14.93	0.483	0.384
Wonderland	17.15	0.550	0.292	16.64	0.574	0.325	15.90	0.510	0.344
Bolt3D	21.54	0.747	0.234	-	-	-	-	-	-
Ours	21.79	0.752	0.219	20.09	0.583	0.313	19.24	0.570	0.336



Figure 6: **Image-to-3DGS Generation.** We visualize five views from generated 3DGS scenes.

humans, animals, and both realistic and imaginative content. We synthesize six camera trajectories for each image (3D) or video (4D), yielding 354,186 videos for 3D and 44,268 videos for 4D.

Baselines. We compare our method with the state-of-the-art approaches for generative single-image-to-3D reconstruction, i.e., ZeroNVS (Sargent et al., 2024), ViewCrafter (Yu et al., 2024b), Wonderland (Liang et al., 2025a), and Bolt3D (Szymanowicz et al., 2025b). Note that *no source code is available* to evaluate these methods on our out-of-distribution evaluation set, hence we mainly rely on reported quantitative comparisons from the papers.

Evaluation. We follow previous works to evaluate our model on the task of single-image to 3D using RealEstate10K (Zhou et al., 2018), DL3DV (Ling et al., 2024b), and Tanks and Temples (Knapitsch et al., 2017). We follow the evaluation protocol outlined in Wonderland (Liang et al., 2025a) and Bolt3D (Szymanowicz et al., 2025b). We evaluate the performance using standard reconstruction metrics, i.e., PSNR, SSIM, and LPIPS.

6.2 MAIN RESULTS

Quantitative results. We show quantitative evaluations for single image-to-3D in Tab. 1. Our method outperforms previous works on all benchmarks across all metrics. Hence, the main improvements to further boost the quality lie in the development of stronger video generative models, as our reconstructions will directly benefit from them. We provide additional quantitative comparisons on the *Lyra* dataset in Appendix C.

Qualitative results. We visualize novel view renderings from our generated 3DGS scenes in Fig. 1 and Fig. 6; video results will be released upon acceptance. Our method generates high-quality novel view content for 3D/4D scenes in unseen regions while maintaining consistency with the input image/video. We visualize qualitative comparisons on the *Lyra* dataset in Appendix C.



Figure 7: **Ablations.** We visualize ablation results by rendering the same extreme novel viewpoint after image-to-3DGS generation; depth visualizations and ablations are provided in Appendix C.4.

6.3 ABLATIONS

In Tab. 2, we motivate our design choices by ablating key components on out-of-distribution diverse prompts. We compare 3DGS renderings with videos generated by our camera-controlled video diffusion model.

Real data only. Instead of using self-distillation, we train a model using only real multi-view datasets, i.e., RealEstate10K (Zhou et al., 2018) and DL3DV (Ling et al., 2024b). These datasets are commonly used in previous works, but lead to limited generalizability to out-of-distribution scenes.

Self-distillation + real data. Perhaps surprisingly, joint training with self-distillation and real data does not improve over using only self-distillation without real data. This confirms that self-distillation is diverse and consistent enough to learn a reconstruction model.

No depth loss. Depth loss prevents flat geometry and even slightly improves image-based metrics. We visualize depth maps in Appendix C.4.

No opacity pruning. We keep all Gaussians instead of pruning them by opacity. Learning to prune most of the 3D Gaussians makes the output representation more compact and slightly improves visual quality. Rendering at $H = 704$, and $W = 1280$ takes 18ms with pruning vs. 30ms without pruning, a $1.67\times$ speed up.

No LPIPS loss. Adding the LPIPS loss improves the robustness to input inconsistencies and enhances the preservation of high-frequency details.

No multi-view fusion. A naive implementation would be to generate 3D Gaussians for each camera trajectory independently and then fuse the 3D Gaussians into one point cloud. However, we observe significant advantages by learning the multi-view fusion from different trajectories with our reconstruction blocks, where each token attends to the others.

No Mamba-2. We replace our joint Transformer/Mamba-2 with Transformer-only blocks and observe slightly lower quality. One forward pass with $V = 6$, $L = 121$, $H = 704$, and $W = 1280$ takes 3213ms with joint blocks in comparison to 20922ms with Transformer-only blocks, a $6.5\times$ speedup.

No latent-based 3DGS. Previous works operating in the pixel space, such as BTimer (Liang et al., 2025b), are restricted to 12 input frames, while we can take up to 726 input frames. Consequently, operating the 3DGS decoder in the pixel space instead of the latent space leads to out-of-memory.

7 CONCLUSION

In this work, we propose *Lyra*, a novel 3D and 4D generation framework relying only on a single image or video input. Instead of collecting multi-view datasets, we introduce camera-controlled video diffusion models as teachers within a self-distillation framework for a student 3DGS decoder. Our 3DGS decoder operates in the latent space of the video model and directly reconstructs 3D Gaussians without any post-processing or optimization. This design enhances generalizability and coverage across diverse scenes, while our dynamic extension demonstrates the feasibility of 4D generation from monocular video. Currently, the scale and consistency of our generated scenes are

Table 2: **Ablation study on *Lyra* dataset.**

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours	24.77	0.837	0.224
Data			
real data only	19.08	0.659	0.413
self-distill. + real data	24.74	0.823	0.236
Loss			
w/o depth loss	24.31	0.811	0.247
w/o opacity pruning	24.55	0.820	0.237
w/o LPIPS loss	23.74	0.766	0.370
Architecture			
w/o multi-view fusion	17.73	0.632	0.446
w/o Mamba-2	24.58	0.818	0.241
w/o latent 3DGS		OOM	

bounded by the capacity of our camera-controlled video diffusion model. Hence, further improving the video diffusion model will enable large-scale, consistent scene synthesis. Moreover, investigating the adaptation of auto-regressive techniques (Chen et al., 2024a) into our framework for large-scale generation is an interesting direction for future work. Lastly, modelling motion and tracking information within the reconstruction network, as in concurrent work (Lin et al., 2025), could improve visual motion quality.

8 ETHICS STATEMENT

Like all generative AI technologies, there are risks of misuse of our method, including producing misleading or synthetic 3D/4D content. While such risks exist, the main goal of this research is to contribute to the fields of simulation, robotics, and embodied AI by providing scalable and controllable tools for data generation. In this sense, we expect our most significant contributions to be made through the following capabilities and insights.

- **The ability to enable realistic simulation for embodied agents:** We provide controllable and physically consistent camera motion within generated 3D and 4D environments to support training and evaluating agents that must perceive, navigate, and interact with complex scenes.
- **Improved scalability for data engines:** Our method removes the need for multi-view capture and expensive per-scene optimization, and helps produce high-fidelity, customizable scenes at scale, which is particularly suitable for research in robotics, reinforcement learning, and closed-loop simulation.
- **Better technical understanding:** As an academic study, this work contributes a foundational case study of how camera control interacts with spatiotemporal generative models, which will benefit research in computer vision, graphics, and embodied AI.

While we caution against deceptive or wrongful use of this technology—where misleading and/or unverifiable media could be created with realistic generative outputs—we believe that the adoption of safeguards (e.g., provenance tracking, dataset documentation, and good evaluation practices) will be useful in ensuring that generative models advance science and society in positive ways.

9 REPRODUCIBILITY STATEMENT

To fully reproduce our results and accelerate future research in this area, our training and inference code, model weights, and data for both 3D and 4D generation will be released upon acceptance.

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A ADDITIONAL DETAILS OF VIDEO DIFFUSION MODEL

This section provides additional information on the GEN3C (Ren et al., 2025) video diffusion model backbone introduced in Sec. 3.1 and how we further improve its 3D consistency.

Conservative mask refinement. A key limitation of GEN3C’s forward warping is background leakage, where occluded object parts in the source view (e.g., lion body in Fig. 8) become visible in novel viewpoints but are not properly indicated in the disocclusion mask. This leads to incomplete foreground completion by the video diffusion model, as shown in Fig. 8 (second and fourth columns).

To address this, we adopt a conservative strategy: since the geometry of occluded regions in $\mathbf{P}^{t,v}$ is unknown, we assume all such areas are occupied and derive refined disocclusion masks accordingly. This allows the video model to reason about potentially visible foreground regions that were missed by standard point-based rendering.

Our approach constructs a triangular mesh from camera-space points $\mathbf{P}_c^{t,v} = \mathbf{C}^t \mathbf{P}^{t,v}$ by connecting spatially adjacent pixels, similar to Hu et al. (2021):

$$\mathcal{M} = \bigcup_{(u,v)} \{ \triangle(\mathbf{p}_{u,v}, \mathbf{p}_{u+1,v}, \mathbf{p}_{u,v+1}), \triangle(\mathbf{p}_{u+1,v}, \mathbf{p}_{u+1,v+1}, \mathbf{p}_{u,v+1}) \} \quad (2)$$

where $\mathbf{p}_{u,v}$ denotes the 3D point at pixel (u, v) in $\mathbf{P}_c^{t,v}$. This surface mesh serves as a boundary between visible and invisible regions. For each rendered pixel, we compare the standard point-based depth $\mathbf{D}^{t,v}$ with the mesh-interpolated depth $\mathbf{D}_{\mathcal{M}}^{t,v}$ obtained via ray-surface intersection:

$$\mathbf{M}^{t,v}(u, v) = \begin{cases} 0 & \text{if } \mathbf{D}_{\mathcal{M}}^{t,v}(u, v) < \mathbf{D}^{t,v}(u, v) - \epsilon \\ \mathbf{M}_{\text{orig}}^{t,v}(u, v) & \text{otherwise} \end{cases}, \quad (3)$$

where $\mathbf{M}_{\text{orig}}^{t,v}$ denotes the original disocclusion mask from forward warping and ϵ is a small tolerance factor. When the mesh surface is closer than the original points, we conservatively mask out these regions as potential foreground disocclusions. This produces more reliable guidance for video generation, especially under large viewpoint changes, as shown in Fig. 8 (third and fifth columns).

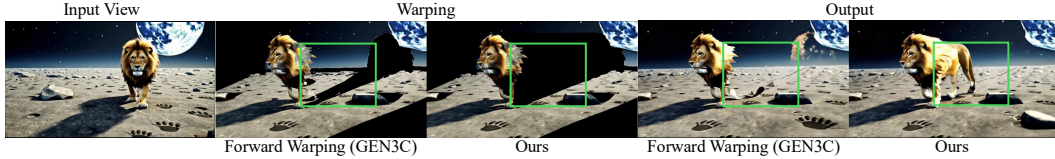


Figure 8: **Rendering function comparison.** Our model builds upon GEN3C (Ren et al., 2025) that uses forward warped images as camera control conditioning. We visualize forward warping (left) vs. our conservative rendering function (right). The highlighted region is incorrectly filled with background pixels in forward warping but properly masked in ours, enabling correct completion.

B MODEL DETAILS

Progressive training. Directly training the 3DGS decoder on 6 trajectories of 121 frames at 704×1280 is expensive, hence we train our network progressively, as shown in Tab. 3. Furthermore, we use the final static pre-trained model to initialize our dynamic model. The total training time takes 6 days with 8 NVIDIA A100 80 GB GPUs. We use gsplat (Ye et al., 2025) as the 3DGS implementation.

Subsampling Gaussians. Generating pixel-aligned 3D Gaussians scales linearly with each spatial and temporal resolution. Our generated scenes represent 726 frames at 704×1280 resolution, leading to 654,213,120 per-pixel Gaussians. Hence, we only generate one 3D Gaussian per 8×8 spatial neighborhood, reducing the total number of 3D Gaussians by a factor of 64, i.e., 10,222,080. Moreover, our opacity-based pruning further reduces the number to 2,044,416, making the output more compact.

Joint Transformer-Mamba blocks. We use the reconstruction block design introduced in Long-LRM (Ziwen et al., 2024), i.e., one block consists of one Transformer layer followed by seven

Table 3: **Progressive training setup.**

Stage	$H \times W$	L	V	S	B	Steps
Static						
1	176×320	17	1	17	4	10k
2	176×320	49	1	49	4	2.5k
3	352×640	49	1	49	2	2.5k
4	704×1280	49	1	49	1	2.5k
5	704×1280	121	1	9	1	57.5k
6	704×1280	121	1-6	9	1	7k
Dynamic						
7	704×1280	121	6	12	1	10k

Mamba-2 layers. We repeat the block twice to get 16 layers with 512 hidden dimensions. Note that our network is significantly smaller than the one used in Long-LRM (Ziwen et al., 2024) or Wonderland (Liang et al., 2025a), which use 24 layers with 1024 hidden dimensions. While pure Transformers are typically more powerful, they are significantly slower in training and inference than Mamba2. Since we trained both configurations within the same training budget, we hypothesize that transformer-only blocks would need considerably more training iterations to fully catch up in rendering quality.

Camera encoding. We use Plücker embeddings for camera encoding, following prior works (Kant et al., 2024; Zhang et al., 2024e). The embeddings are computed at full spatial and temporal resolution and then projected into the video latent space using the pre-trained RGB encoder. Since the RGB encoder expects 3 input channels but Plücker embeddings are 6-dimensional, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins. The resulting latent encodings are then concatenated along the channel dimension.

Time encoding. Similarly, to make the 1-dimensional time input compatible with the 3-channel input expected by the RGB encoder, we concatenate a 2-dimensional sinusoidal embedding to the original time value along the channel dimension. We then replicate these values across the spatial dimensions before encoding the time into the video latent space. This approach allows us to independently encode both the source and target times into the compressed latent representation.

Dynamic data augmentation. We visualize our dynamic data augmentation procedure for two example camera trajectories in Fig. 9. The procedure creates pairs of supervision views that cover the scene from the same motion state but different extreme viewpoints.

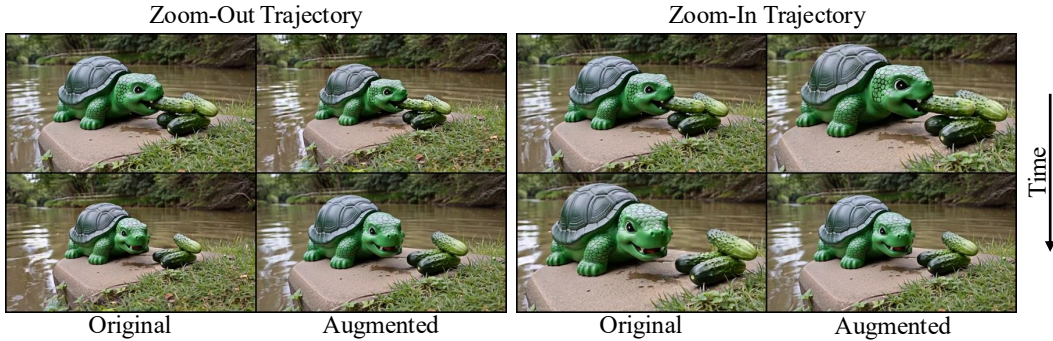


Figure 9: **Dynamic data augmentation videos.** We augment the supervision data with a motion-reversed video, ensuring that each timestep is observed from the full spatial coverage, thereby preventing low opacity artifacts in the early timesteps. We show two example trajectories, i.e., zoom-out and zoom-in, and visualize their corresponding augmented videos. The augmented videos are flipped in their camera motion.



Figure 10: **Image-to-3DGS Comparison.** We compare our method with BTimmer (GEN3C) and visualize five views from generated scenes. We observe significantly fewer artifacts and higher fidelity.

C ADDITIONAL EVALUATION

C.1 ADDITIONAL BASELINE DETAILS

We additionally compare our model against BTimmer (Liang et al., 2025b), a recent approach for joint 3D and 4D reconstruction from posed images or videos. We reached out to the authors of BTimmer to conduct experiments for us using the same scenes as *Lyra*. Since BTimmer is purely regression-based, we integrate it with our GEN3C (Ren et al., 2025) video diffusion backbone. Specifically, we use the same camera trajectories as in our model and provide RGB-decoded videos from GEN3C as input to BTimmer. BTimmer operates in pixel space and requires 12 input images. However, as discussed in the main paper, pixel-space models such as BTimmer run out of memory when directly using the 726 high-resolution frames generated by GEN3C. To address this, we uniformly subsample GEN3C frames. For static evaluations, we select 12 frames that evenly cover the available viewpoints. For dynamic evaluations, we sample 11 frames to span the viewpoint range and additionally include the bullet-time frame corresponding to the motion being reconstructed. This ensures that the target motion state is always present as a reference. We denote this integrated baseline as BTimmer (GEN3C), i.e., BTimmer combined with GEN3C.

C.2 ADDITIONAL STATIC 3D EVALUATION

For the task of image to 3DGS generation, we show qualitative comparisons with BTimmer (GEN3C) in Fig. 10 and qualitative results in Tab. 4a. We observe higher quality and less artifacts for our method.

C.3 DYNAMIC 3D EVALUATION

BTimmer (GEN3C) comparison. For dynamic 3D evaluation, we compare our method with BTimmer (GEN3C), as discussed in Sec. C.1, on 100 out-of-distribution videos from our dynamic *Lyra* dataset. We crop the video to 512×512 for fair comparisons, as BTimmer was mainly trained on that resolution. We show results in Tab. 4b and observe that our method significantly outperforms BTimmer (GEN3C).

Table 4: Quantitative results on static and dynamic *Lyra* datasets.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BTimer (GEN3C)	16.32	0.580	0.427	BTimer (GEN3C)	20.29	0.687	0.315
Ours	24.92	0.834	0.183	Ours	23.07	0.779	0.231

(a) Comparison on static *Lyra* dataset. (b) Comparison on dynamic *Lyra* dataset.

C.4 DEPTH VISUALIZATION

We visualize depths from our generated 3DGS in Fig. 11. Using depth supervision prevents flat geometries without sacrificing visual quality.

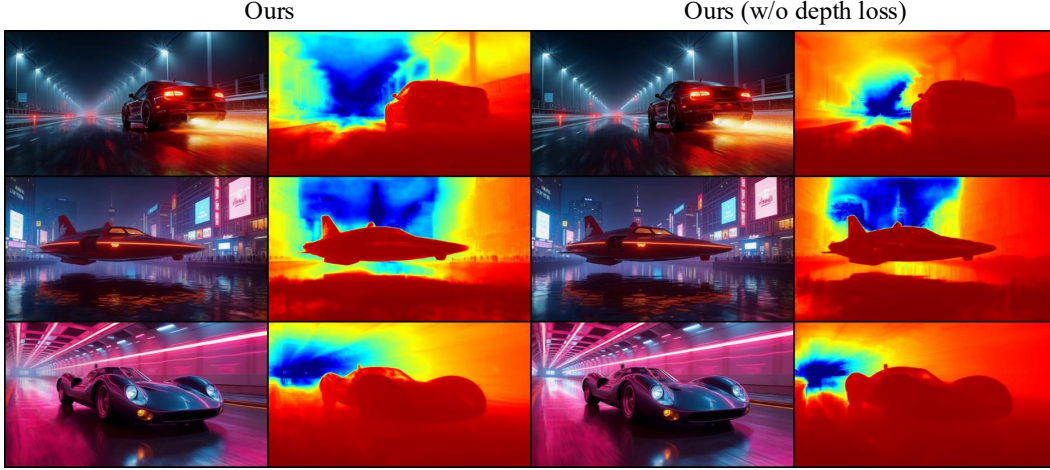


Figure 11: **Depth loss ablation.** We visualize 3DGS renderings and corresponding depth maps. Using the depth loss as additional supervision prevents flat geometries without sacrificing visual quality.

D ROBOT SIMULATION

One of our primary motivations for building a diverse 3D generator is its ability to synthesize simulation environments for autonomous agents. To this end, we develop an end-to-end pipeline: we first generate 3D scenes from text with our model, export the 3D Gaussians as a .ply file, and convert the .ply files into a .usdz format. We use 3DGUT (Wu et al., 2025a) and its export function to create .usdz files. The resulting .usdz file can then be imported into the NVIDIA Isaac robot development platform as a physically based virtual environment. AI-based robots can be trained and tested under diverse conditions within these environments. Early demo results will be released upon acceptance; we include a screenshot of the interface in Fig. 12.

E RELATED WORK

In this section, we will discuss additional related work. We highlight the differences to previous works CAT3D (Gao et al., 2024b) and Bolt3D (Szymanowicz et al., 2025b) in Fig. 13.

3D generation. Early approaches to 3D generation primarily focused on single object categories, extending GANs into the 3D domain by leveraging neural renderers as an inductive bias (DeVries et al., 2021; Chan et al., 2022; Or-El et al., 2022; Schwarz et al., 2022; Bahmani et al., 2023b). With the introduction of CLIP-based supervision (Radford et al., 2021), the field advanced toward more flexible and diverse asset creation, enabling both text-driven generation and editing (Chen et al., 2018; Jain et al., 2022; Sanghi et al., 2022; Jetchev, 2021; Gao et al., 2023; Wang et al., 2022). More recently, diffusion models have significantly improved quality by replacing CLIP guidance with Score Distillation Sampling (SDS) (Poole et al., 2023; Wang et al., 2023b; Lin et al., 2023a; Chen et al.,

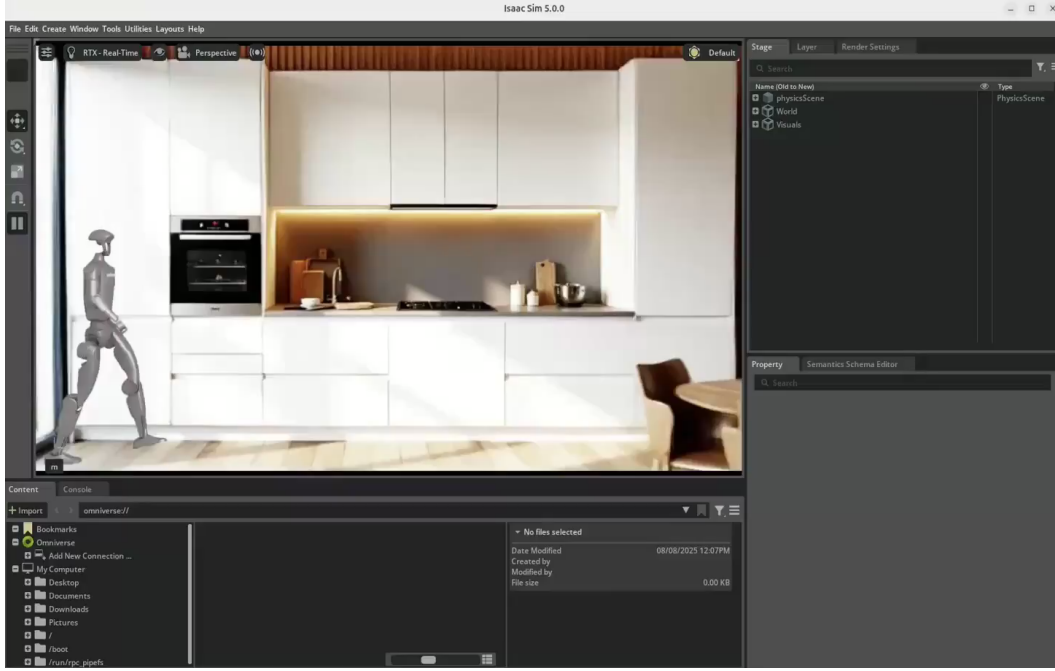


Figure 12: **Robot Simulation.** We visualize a frame of a robot simulation within the Isaac Sim 5.0 simulation framework that takes a generated 3DGS scene from our method as input.

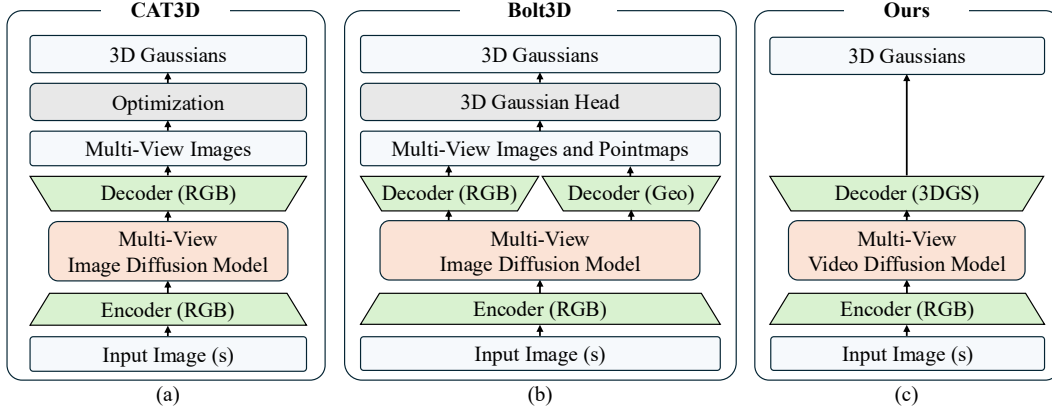


Figure 13: **Approaches for 3D Generation.** CAT3D (Gao et al., 2024b) proposed a multi-view image diffusion model that outputs images from novel viewpoints; subsequently, the images are reconstructed into 3D Gaussians using optimization. Bolt3D (Szymanowicz et al., 2025b) fine-tunes CAT3D to output pointmaps using a geometry autoencoder; instead of relying on optimization, the 3D scene is predicted with a feed-forward 3D Gaussian head. In contrast, our work builds upon a pre-trained camera-controlled video diffusion model and *directly* decodes the multi-view video latents into 3D Gaussians.

2023b; Liang et al., 2023; Wang et al., 2023a; Li et al., 2024e; He et al., 2024b; Ye et al., 2024; Liu et al., 2024b; Yu et al., 2023; Katzir et al., 2024; Lee et al., 2024a; Sun et al., 2024a).

To further enhance structural coherence, several methods enforce multi-view consistency by generating scenes from different perspectives (Lin et al., 2023b; Liu et al., 2023; Shi et al., 2024; Feng et al., 2024a; Liu et al., 2024a; Kim et al., 2023; Voleti et al., 2024; Höllein et al., 2024; Tang et al., 2024c; Gao et al., 2024b; Wang et al., 2025f; Kant et al., 2025; Yuan et al., 2025; Ren et al., 2024b), while others adopt iterative inpainting as a strategy for scene synthesis (Höllein et al., 2023; Shriram et al., 2024). Another line of work explores lifting 2D images into 3D using NeRF (Mildenhall et al., 2020), 3D Gaussian Splatting (Kerbl et al., 2023), or mesh-based representations coupled with diffusion

models (Chan et al., 2023; Tang et al., 2023a; Gu et al., 2023; Liu et al., 2024c; Yoo et al., 2023; Tewari et al., 2023; Qian et al., 2024b; Long et al., 2024; Wan et al., 2024; Szymanowicz et al., 2023; Lu et al., 2024).

Feed-forward 3D models. Complementary efforts investigate fast, feed-forward techniques that directly predict 3D content from images or text inputs (Hong et al., 2024; Li et al., 2024b; Xu et al., 2024e;d; Zhang et al., 2024a; Han et al., 2024; Jiang & Wang, 2024; Xie et al., 2024a; Tang et al., 2024b; Tochilkin et al., 2024; Qian et al., 2024a; Szymanowicz et al., 2024; 2025a; Liang et al., 2025a; Szymanowicz et al., 2025b; Schwarz et al., 2025; Yang et al., 2025; Zhang et al., 2025a). However, in contrast to our method, these approaches remain limited to producing static 3D scenes. Other methods focus on specific scenes such as faces (Kirschstein et al., 2025). Concurrent work (Liang et al., 2025b; Xu et al., 2025) tackles real scenes but can not handle diverse generated scenes or large viewpoint changes.

4D generation. Recent years have seen rapid progress in 4D generation, i.e., dynamic 3D scene synthesis. Many approaches leverage input text prompts or images as guidance. Following the introduction of large-scale generative models for this task (Singer et al., 2023), subsequent works have made notable advances in both visual fidelity and motion quality (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024b; Gao et al., 2024a; Yang et al., 2024a; Jiang et al., 2024b; Miao et al., 2024; Li et al., 2024a; Zhang et al., 2024g; Yuan et al., 2024; Jiang et al., 2024a).

While text conditioning is widely used, alternative methods aim to lift 2D images or videos into dynamic 3D scenes (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024b; Ling et al., 2024a; Gao et al., 2024a; Zeng et al., 2024b; Chu et al., 2024; Wu et al., 2024; Yang et al., 2024b; Wang et al., 2024c; Feng et al., 2024b; Sun et al., 2024c; Zhang et al., 2024c; Ren et al., 2024a; Lee et al., 2024b; Li et al., 2024c; Van Hoorick et al., 2024; Uzolas et al., 2024; Chai et al., 2024; Liang et al., 2024; Li et al., 2024d; Xie et al., 2024b; Li et al., 2025a; Wang et al., 2025e; Zhu et al., 2025; Zhou et al., 2025; Hu et al., 2025; Park et al., 2025). Other efforts explore incorporating physics priors into generation pipelines (Lin et al., 2024; Huang et al., 2024b; Zhang et al., 2024f; Kiray et al., 2025), or enhancing motion controllability with template-driven approaches (Zhang et al., 2024d; Sun et al., 2024b; Chen et al., 2024b). A parallel direction emphasizes compositional and interactive 4D generation (Bahmani et al., 2024a; Xu et al., 2024b; Cao et al., 2024; Yu et al., 2024a; Zeng et al., 2024a; Zhu et al., 2024).

Another strand of research extends 3D GAN frameworks into the 4D setting by training directly on 2D video data (Bahmani et al., 2023a; Xu et al., 2023). However, such methods are typically limited by small-scale, single-category datasets, which constrain generalization. Moreover, most existing approaches remain object-centric, often neglecting background elements. As a result, their overall visual fidelity lags behind the high photorealism achieved by recent video generation models, which our method leverages.

Camera-conditioned video models. Recent progress has focused on incorporating camera control into video diffusion models. The pioneering work MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models (Chen et al., 2023a; Blattmann et al., 2023) with extrinsic matrices. Subsequent efforts (He et al., 2024a; Xu et al., 2024c; Kuang et al., 2024; Ju et al., 2025; He et al., 2025; Li et al., 2025c; Wang et al., 2025c; 2024d; Lei et al., 2025; Liu et al., 2025) enhanced conditioning by representing cameras using Plücker coordinates, while others use follow MotionCtrl to simply use poses flattened (Wang et al., 2025b). Another line of research (Hu et al., 2024; Xiao et al., 2024; Ling et al., 2024c; Hou et al., 2024) enabled camera motion control without introducing additional trainable parameters. Despite these advances, all of the above rely on U-Net-based architectures. Other approaches use low-rank approaches for camera control (Zhang et al., 2025b).

Several recent works continue to extend U-Net-based approaches for camera control (Zhao et al., 2024; Xu et al., 2024a; Zheng et al., 2024a; Zhang et al., 2024b; Yu et al., 2024b; Lei et al., 2024), while others begin exploring diffusion transformers. Cheong et al. (Cheong et al., 2024) propose one such transformer-based framework, though its scene and visual quality remain limited. DimensionX (Sun et al., 2024d) introduces joint space-time control in diffusion transformers, but relies on pre-defined, non-continuous camera trajectories. Other works (Huang et al., 2024a; Mai et al., 2025; Jiang et al., 2025b) instead explore pose and geometry estimation with a video DiT. CAT4D (Wu et al., 2025b) proposes a multi-view video diffusion model derived from fine-tuning a multi-view backbone. Large-

scale synthetic data generation further boosts performance, as demonstrated by SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a), which achieve strong results on camera-controlled video generation using transformer-based architectures. Finally, 4DiM (Watson et al., 2024) trains a space-time diffusion model from scratch for novel view synthesis from a single input image. Another line of work edits camera movements of videos with underlying motion using camera retargeting (Seo et al., 2025; Ma et al., 2025). One popular direction has been to use a spatio-temporal cache, e.g., using point clouds or optical flow, to condition the camera-controlled video generation (Wu et al., 2025c; Ren et al., 2025; Wang et al., 2025d; Cao et al., 2025; YU et al., 2025; Jin et al., 2025; Xing et al., 2025; Yan et al., 2025; Li et al., 2025b; Gu et al., 2025).